ECE 276A Project 2: LiDAR-Based SLAM

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I. Introduction

Indoors there is no GPS, and odometry drifts: on dataset 20 it ends 2.77 m from the start after a 95.9 m drive. SLAM (Simultaneous Localization and Mapping) estimates the path and the map together, so that seeing the same wall twice corrects the path.

The system has four stages for a differential-drive robot with encoders, an IMU, a 2-D LiDAR and a Kinect [1] (Fig. 1): (1) encoder and IMU odometry, (2) ICP scan matching, (3) pose graph optimization with loop closures in GTSAM [3], and (4) occupancy grid and floor texture mapping with the optimized poses. The final path ends 0.12 m (dataset 20) and 0.22 m (dataset 21) from the start, compared to 2.7 to 3.3 m for odometry or ICP alone.

II. Problem Formulation

Pose. $\mathbf{x} = (x, y, \theta)$ with $\theta \in [-\pi, \pi)$, and matrix form

$$T(\mathbf{x}) = \begin{bmatrix} R(\theta) & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix}, \qquad (1)$$

which maps a body point to the world: $\mathbf{p}_W = R\mathbf{p}_B + \mathbf{t}$. The pose of j relative to i is $T_i^{-1} T_j$. For scan k, $T_k = T(\mathbf{x}_k)$ has rotation $R_k = R(\theta_k)$ and translation $\mathbf{t}_k$. The unknowns are the path $\{\mathbf{x}_0, \dots, \mathbf{x}_K\}$ (one pose per LiDAR scan), the occupancy map m and the floor color map c.

Motion model. With speed v_t, yaw rate ω_t, time step τ_t:

$$\theta_{t+1} = \theta_t + \omega_t \tau_t, \quad \begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} x_t \\ y_t \end{bmatrix} + v_t \tau_t \begin{bmatrix} \cos\theta_{t+1} \\ \sin\theta_{t+1} \end{bmatrix}. \quad (2)$$

Registration. For point sets $\{\mathbf{p}_k\}$, $\{\mathbf{q}_k\}$ with known pairs,

$$(R^*, \mathbf{t}^*) = \arg\min_{R,\,\mathbf{t}} \sum_{k=1}^{N} \|\mathbf{q}_k - (R\mathbf{p}_k + \mathbf{t})\|^2 \quad \text{subject to} \quad R^\top R = I,\ \det R = 1. \qquad (3)$$

ICP solves the same problem when the pairs are unknown.

Pose graph. Each edge $(i, j) \in \mathcal{E}$ carries a measured relative pose $\mathbf{u}_{ij} = (\mathbf{u}_{ij}^{xy}, u_{ij}^{\theta})$ (pose j seen from pose i, from ICP) with covariance Σ_{ij} and information matrix $\Omega_{ij} = \Sigma_{ij}^{-1}$. Composing pose i with the measurement predicts pose j:

$$f(\mathbf{x}_i, \mathbf{u}_{ij}) = \begin{bmatrix} \mathbf{t}_i + R(\theta_i)\, \mathbf{u}_{ij}^{xy} \\ \theta_i + u_{ij}^{\theta} \end{bmatrix}, \qquad \mathbf{e}_{ij} = \mathbf{x}_j - f(\mathbf{x}_i, \mathbf{u}_{ij}), \qquad (4)$$

with the angle entry of $\mathbf{e}_{ij}$ wrapped to $[-\pi, \pi)$. The estimate minimizes

$$\mathbf{x}^* = \arg\min_{\mathbf{x}} \ \mathcal{J}(\mathbf{x}), \qquad \mathcal{J}(\mathbf{x}) = \tfrac{1}{2} \sum_{(i,j)\in\mathcal{E}} \mathbf{e}_{ij}^\top \Omega_{ij}\, \mathbf{e}_{ij}. \quad (5)$$

GTSAM's Pose2 factor writes the same error rotated into the frame of the prediction, which leaves $\mathcal{J}$ unchanged because $\sigma_x = \sigma_y$ for every edge.

Occupancy grid. Each cell m_i holds the log-odds $\lambda_i = \log \frac{p(m_i=1|z_{1:t})}{p(m_i=0|z_{1:t})}$, with $\lambda_i = 0$ meaning unknown (50 %).

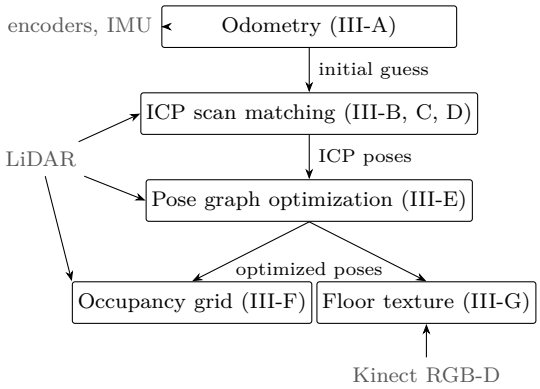


Fig. 1. Pipeline. Labels give the subsections of Section III.

III. Technical Approach

Table I gives the input and output of each function in the code.

A. Odometry

Step 1. Distance per tick, wheel diameter $D = 0.254$ m, 360 ticks per turn: $s = \pi D/360 \approx 0.0022$ m.

Step 2. Side travel from the counts FR, FL, RR, RL since the last reading:

$$d_R = \tfrac{FR+RR}{2}\, s, \qquad d_L = \tfrac{FL+RL}{2}\, s. \qquad (6)$$

Step 3. The center of a differential-drive robot moves by the average of the two sides, so with τ the encoder time step,

$$l = \tfrac{d_R+d_L}{2}, \qquad v = \tfrac{l}{\tau}. \qquad (7)$$

TABLE I

Functions in main.py and icp.py. N = number of points, K = number of LiDAR scans (4962 for dataset 20).

Function	Input	Returns
odometry(d)	sensor data	(4956, 3) array of (x, y, θ), one per encoder reading
lidar_points(d, k)	scan number k	(N, 2) points of scan k in the body frame (meters)
kabsch(P, Q)	two (N, d) arrays, row k paired with row k	R, t, rmsd: best rotation, translation, fit error
icp(source, target, T_initial)	two clouds, no pairs, a starting guess	T, error: $(d{+}1)\times(d{+}1)$ transform source→target, mean pair distance
scan_matching(d, odom)	data, odometry poses	(K, 3) poses, one per LiDAR scan
match_scans(d, poses_T, i, j, …)	two scan numbers	T_relative, inlier_ratio: pose of j relative to i, share of points within 10 cm
pose_graph(d, icp_poses)	data, (K, 3) ICP poses	(K, 3) optimized poses
occupancy_map(d, poses)	data, (K, 3) poses	map_info, grid, snapshots: 1600×1600 log-odds grid + 4 copies over time
camera_to_body()	none	3×3 rotation, Kinect frame → body frame
texture_map(d, dataset, poses)	data, poses	map_info, texture, snapshots: 1600×1600×3 color image + 4 copies

Step 4. Yaw rate: the wheels give $\omega = (d_R - d_L)/(b\,\tau)$ with track width b, but wheels slip in turns, so ω is taken from the IMU z-axis gyro at the nearest IMU time stamp.
Step 5. Integrate with the Euler step (2), assuming v and ω constant over each τ.

B. Kabsch algorithm (known pairs)

Step 1. Objective (3): $E(R,\mathbf{t}) = \sum_k \|\mathbf{q}_k - R\mathbf{p}_k - \mathbf{t}\|^2$.
Step 2. Optimal translation. Setting the gradient to zero,

$$\frac{\partial E}{\partial \mathbf{t}} = -2\sum_k (\mathbf{q}_k - R\mathbf{p}_k - \mathbf{t}) = 0 \;\Rightarrow\; \mathbf{t}^* = \bar{\mathbf{q}} - R\bar{\mathbf{p}}, \quad (8)$$

with centroids $\bar{\mathbf{p}} = \frac{1}{N}\sum_k \mathbf{p}_k$, $\bar{\mathbf{q}} = \frac{1}{N}\sum_k \mathbf{q}_k$. This is why centroids are used: the best translation always moves the source centroid onto the target centroid, so after centering both sets the translation drops out and the rotation can be solved alone.
Step 3. Centered points $\mathbf{p}'_k = \mathbf{p}_k - \bar{\mathbf{p}}$, $\mathbf{q}'_k = \mathbf{q}_k - \bar{\mathbf{q}}$. Substituting $\mathbf{t}^*$ and expanding, with $\|R\mathbf{p}'_k\| = \|\mathbf{p}'_k\|$:

$$E(R) = \sum_k \left(\|\mathbf{q}'_k\|^2 + \|\mathbf{p}'_k\|^2\right) - 2\sum_k \mathbf{q}'^{\top}_k R\,\mathbf{p}'_k. \quad (9)$$

The first sum does not depend on R, so minimizing E means maximizing the last sum.
Step 4. Trace form. Since $\mathbf{q}'^{\top} R\mathbf{p}' = \mathrm{tr}(R\,\mathbf{p}'\mathbf{q}'^{\top})$,

$$\begin{aligned} \textstyle\sum_k \mathbf{q}'^{\top}_k R\,\mathbf{p}'_k &= \mathrm{tr}(RH), \\ H &= \textstyle\sum_k \mathbf{p}'_k\mathbf{q}'^{\top}_k = P'^{\top}Q' \in \mathbb{R}^{d\times d}. \end{aligned} \quad (10)$$

Step 5. SVD. Every real matrix factors as

$$\begin{aligned} H = U\Sigma V^{\top}, \quad U^{\top}U = V^{\top}V = I, \\ \Sigma = \mathrm{diag}(\sigma_1 \ge \cdots \ge \sigma_d \ge 0), \end{aligned} \quad (11)$$

i.e. a rotation (or reflection), a scaling along the axes, and another rotation. σ_i are the singular values.
Step 6. Bound. By the cyclic property of the trace,

$$\begin{aligned} \mathrm{tr}(RH) &= \mathrm{tr}(RU\Sigma V^{\top}) = \mathrm{tr}(\Sigma\, V^{\top}RU) \\ &= \textstyle\sum_i \sigma_i M_{ii}, \qquad M = V^{\top}RU. \end{aligned} \quad (12)$$

M is orthogonal, so $|M_{ii}| \le 1$ and $\mathrm{tr}(RH) \le \sum_i \sigma_i$. Equality holds for $M = I$, so $R^* = VU^{\top}$.
Step 7. Reflection fix. If $\det(VU^{\top}) = -1$, $VU^{\top}$ is a mirror, not a rotation. The best proper rotation flips the axis with the smallest σ_d:

$$R^* = V\,\mathrm{diag}\big(1,\ldots,1,\det(VU^{\top})\big)U^{\top}, \quad (13)$$

which the code does by negating the last row of $V^{\top}$. A self-test rotates a random cloud by 40° and recovers R and $\mathbf{t}$ exactly.

C. ICP (unknown pairs)

Step 1. Start from T^0 and build a KD-tree on the target once.
Step 2. Transform: $\tilde{\mathbf{p}}_k = R^n\mathbf{p}_k + \mathbf{t}^n$.
Step 3. Correspondence (nearest neighbor):

$$j(k) = \arg\min_j \|\mathbf{q}_j - \tilde{\mathbf{p}}_k\|.$$

Step 4. Outlier rejection:

$$\mathcal{I} = \{k : \|\mathbf{q}_{j(k)} - \tilde{\mathbf{p}}_k\| < d_{\max}\}.$$

Step 5. Update with Kabsch on the kept pairs:

$$(R^{n+1}, \mathbf{t}^{n+1}) = \mathrm{Kabsch}\big(\{\mathbf{p}_k, \mathbf{q}_{j(k)}\}_{k\in\mathcal{I}}\big).$$

Step 6. Error and stopping rule:

$$e^n = \frac{1}{|\mathcal{I}|}\sum_{k\in\mathcal{I}} \|\mathbf{q}_{j(k)} - \tilde{\mathbf{p}}_k\|,$$

$$\text{stop if } |e^{n-1} - e^n| < 10^{-6} \text{ or } n = 50.$$

Convergence. With $E(T, j)$ the error for transform T and pairing j,

$$E(T^n, j^n) \;\ge\; E(T^n, j^{n+1}) \;\ge\; E(T^{n+1}, j^{n+1}), \quad (14)$$

because step 3 picks the closest pairs for the current T and step 5 picks the best T for those pairs. A non-increasing sequence bounded below by 0 converges, so ICP stops, but only at the nearest local minimum.

A far-off T^0 gives wrong pairs from the first iteration, so a good initial guess is required. (With $d_{\max}$ the kept set changes, so the decrease is nearly, not strictly, monotone.)

Warm-up. The partial view is registered onto the full model and the result inverted: every view point has a partner on the model, but the hidden back of the model has none in the view. ICP is started from 12 yaws $(0°, 30°, \ldots, 330°)$, each with the centroids aligned, using every 10th view point; the start with the lowest error is refined with all view points.

D. LiDAR scan matching

Step 1. Ranges to body points: for beam n with range r_n, $\alpha_n = \alpha_{\min} + n\Delta\alpha$ and

$$\mathbf{p}_n = \big(r_n \cos\alpha_n + 0.133,\ r_n \sin\alpha_n\big), \quad r_{\min} < r_n < r_{\max}, \tag{15}$$

where 0.133 m (0.13323 in the code) is the LiDAR offset from the robot center [2]. $S_k = \{\mathbf{p}_n\}$ is the point set of scan k.

Step 2. Relative pose. A wall point has one world position: $T_i\mathbf{p}^{(i)} = T_j\mathbf{p}^{(j)} \Rightarrow \mathbf{p}^{(i)} = (T_i^{-1}T_j)\,\mathbf{p}^{(j)}$. So ICP of scan j onto scan i estimates $T_i^{-1}T_j$.

Step 3. Initial guess from odometry poses O, relative to keyframe kf:

$$T^0 = T_{\text{kf}}^{-1}\, T_{k-1}\, (O_{k-1}^{-1} O_k). \tag{16}$$

$O_{k-1}^{-1}O_k$ is the odometry step in the robot frame, so it can be stacked on the ICP pose T_{k-1} even after the two paths drift apart.

Step 4. Match and compose: $T_{\text{rel}} = \text{ICP}(S_k, S_{\text{kf}}, T^0)$, $T_k = T_{\text{kf}}T_{\text{rel}}$.

Step 5. Scan k becomes the keyframe when $\|\mathbf{t}_{\text{rel}}\| > 0.5\,\text{m}$ or $|\theta_{\text{rel}}| > 0.2\,\text{rad}$. Consecutive scans are only 2 cm apart; ICP under-reports such a small motion in a hallway, and chaining 4962 such steps made the hallways too short.

E. Pose graph optimization

Step 1. Variables: $\mathbf{x} = (\mathbf{x}_0, \ldots, \mathbf{x}_K)$, initialized with the ICP poses.

Step 2. Edges $\mathcal{E}$, each with $\mathbf{e}_{ij}$ from (4):

- prior on $\mathbf{x}_0$: $\mathbf{e}_0 = \mathbf{x}_0 - \bar{\mathbf{x}}_0$ ($\bar{\mathbf{x}}_0$ the first ICP pose), $\sigma = 10^{-3}$;
- odometry $(k-1, k)$: $\mathbf{u}$ = ICP step $T_{k-1}^{-1}T_k$, $\sigma = (0.01, 0.01, 0.005)$;
- fixed-interval loops $(i, i+10)$, ICP with $d_{\max} = 0.5\,\text{m}$;
- proximity loops (i, j), every 20th scan, $\|\mathbf{t}_i - \mathbf{t}_j\| < 2\,\text{m}$ and $j - i \geq 500$, ICP with $d_{\max} = 1.0\,\text{m}$.

A loop edge $\mathbf{u}_{ij} = \text{ICP}(S_j, S_i, T_i^{-1}T_j)$ is kept only if its inlier ratio

$$\rho_{ij} = \frac{1}{N}\sum_k \mathbf{1}\Big[\min_{\mathbf{q}\in S_i} \|U_{ij}\mathbf{p}_k - \mathbf{q}\| < 0.1\Big] \geq 0.6, \tag{17}$$

with U_{ij} the matrix form of $\mathbf{u}_{ij}$. Loop edges use $\sigma = (0.1, 0.1, 0.05)$.

Step 3. Robust cost. With the whitened error $\varepsilon_{ij} = \sqrt{\mathbf{e}_{ij}^\top \Omega_{ij} \mathbf{e}_{ij}}$,

$$\mathcal{J}(\mathbf{x}) = \sum_{\text{prior, odo}} \tfrac{1}{2}\varepsilon_{ij}^2 + \sum_{\text{loop}} \rho_H(\varepsilon_{ij}),$$
$$\rho_H(\varepsilon) = \begin{cases} \tfrac{1}{2}\varepsilon^2, & \varepsilon \leq 1 \\ \varepsilon - \tfrac{1}{2}, & \varepsilon > 1 \end{cases} \tag{18}$$

ρ_H grows linearly for large ε, so a wrong loop pulls with bounded force.

Step 4. Linearize around the current estimate $\mathbf{x}^{(n)}$: $\mathbf{e}_{ij}(\mathbf{x}^{(n)} + \delta\mathbf{x}) \approx \mathbf{e}_{ij} + J_i\,\delta\mathbf{x}_i + J_j\,\delta\mathbf{x}_j$. Differentiating (4), with $\frac{d}{d\theta}R(\theta) = R(\theta)S$ and $S = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$,

$$J_i = \frac{\partial \mathbf{e}_{ij}}{\partial \mathbf{x}_i} = \begin{bmatrix} -I_2 & -R(\theta_i)S\,\mathbf{u}_{ij}^{xy} \\ \mathbf{0}^\top & -1 \end{bmatrix}, \qquad J_j = \frac{\partial \mathbf{e}_{ij}}{\partial \mathbf{x}_j} = I_3. \tag{19}$$

Step 5. Quadratic model and its minimum. Substituting into (5) and setting the gradient with respect to $\delta\mathbf{x}$ to zero gives the normal equations $H\,\delta\mathbf{x} = -\mathbf{b}$, assembled edge by edge:

$$\begin{aligned} H_{ii} &\mathrel{+}= J_i^\top \Omega_{ij} J_i, & H_{ij} &\mathrel{+}= J_i^\top \Omega_{ij} J_j, \\ H_{ji} &\mathrel{+}= J_j^\top \Omega_{ij} J_i, & H_{jj} &\mathrel{+}= J_j^\top \Omega_{ij} J_j, \\ \mathbf{b}_i &\mathrel{+}= J_i^\top \Omega_{ij} \mathbf{e}_{ij}, & \mathbf{b}_j &\mathrel{+}= J_j^\top \Omega_{ij} \mathbf{e}_{ij}. \end{aligned} \tag{20}$$

The prior adds Ω_0 to H_{00} and $\Omega_0\mathbf{e}_0$ to $\mathbf{b}_0$.

Step 6. Levenberg-Marquardt step and update:

$$(H + \mu I)\,\delta\mathbf{x} = -\mathbf{b}, \qquad \mathbf{x}^{(n+1)} = \mathbf{x}^{(n)} + \delta\mathbf{x} \tag{21}$$

(angles wrapped). If $\mathcal{J}$ dropped, μ decreases (toward Gauss-Newton); otherwise the step is rejected and μ increases (toward gradient descent). Repeat until $\delta\mathbf{x}$ is small.

Step 7. Prior and sparsity. Without the prior, shifting or rotating the whole path leaves $\mathcal{J}$ unchanged, so the minimum is not unique and H is singular. Each edge touches 2 poses, so H (14886×14886) has non-zero blocks only on the diagonal and at edge pairs, and is solved by sparse elimination. For one test factor, $\frac{1}{2}\sum(e/\sigma)^2 = 10.5001$ by hand and from factor.error(), and a plain implementation of steps 2 to 6 on a 5-pose test loop matched GTSAM's optimum to 3×10^{-8}.

The small odometry σ is deliberate: with $\sigma = 5\,\text{cm}$ the loop closures shrank the path to 86.5 m (Table III), because sliding along a corridor barely changes scan overlap.

F. Occupancy grid mapping

Step 1. Beams in the world: $\mathbf{p}_W = R_k\mathbf{p} + \mathbf{t}_k$ and LiDAR origin $\mathbf{o} = T_k[0.133, 0, 1]^\top$.

Step 2. Cell index with resolution $r = 0.05\,\text{m}$ and map corner $\mathbf{m}_{\min}$: $\text{cell}(\mathbf{p}) = \lfloor(\mathbf{p} - \mathbf{m}_{\min})/r\rfloor$, on a 1600×1600 grid.

Step 3. Cells on each beam: $\{c_1, \ldots, c_L\} = \text{Bresenham}\big(\text{cell}(\mathbf{o}), \text{cell}(\mathbf{p}_W)\big)$.

Step 4. Bayes' rule for one cell: $p(m \mid z_{1:t}) = \eta\, p(z_t \mid m)\, p(m \mid z_{1:t-1})$.

Step 5. Divide the expression for $m = 1$ by the one for $m = 0$; η cancels:

$$\frac{p(m{=}1 \mid z_{1:t})}{p(m{=}0 \mid z_{1:t})} = g \cdot \frac{p(m{=}1 \mid z_{1:t-1})}{p(m{=}0 \mid z_{1:t-1})}, \quad g = \frac{p(z_t \mid m{=}1)}{p(z_t \mid m{=}0)}. \tag{22}$$

Step 6. Take the log, so the product becomes a sum:

$$\lambda_t = \lambda_{t-1} + \log g. \tag{23}$$

Step 7. Sensor model with 80 % trust ($g = 0.8/0.2 = 4$): the beam passed through $c_1, \ldots, c_{L-1}$, so λ $-{=}$ $\log 4$ (free), and it ended in c_L, so λ $+{=}$ $\log 4$ (occupied). Clip to $[-20, 20]$ so cells can still change. Every 2nd scan is used.

Step 8. Probability: $p(m{=}1) = 1 - \dfrac{1}{1 + e^{\lambda}}$ (the logistic function).

G. Floor texture mapping

For every 5th Kinect image, with the nearest RGB image and pose in time:

Step 1. Depth from disparity d (handout calibration): $dd = -0.00304\, d + 3.31$, $z = 1.03/dd$.

Step 2. Matching RGB pixel for depth pixel (u, v) (column, row):

$$u_{\mathrm{rgb}} = \tfrac{526.37\, u + 16662}{585.051}, \quad v_{\mathrm{rgb}} = \tfrac{526.37\, v + 19276 - 7877.07\, dd}{585.051}. \tag{24}$$

Step 3. Pinhole model. Projection with intrinsics K:

$$z \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad K = \begin{bmatrix} f & 0 & c_u \\ 0 & f & c_v \\ 0 & 0 & 1 \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = z\, K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \tag{25}$$

so $x = (u - c_u)z/f$, $y = (v - c_v)z/f$, with $f = 585.05$, $c_u = 315.84$, $c_v = 242.94$ (optical frame: x right, y down, z forward).

Step 4. Optical to camera frame (x forward, y left, z up): $\mathbf{p}_C = (z, -x, -y)$.

Step 5. Camera to body: $\mathbf{p}_B = R_z(\psi) R_y(\phi) R_x(\gamma)\, \mathbf{p}_C + [0.18, 0.005, 0.36]^{\top}$, with roll $\gamma = 0$, pitch $\phi = 0.36$, yaw $\psi = 0.021$ rad. Each factor keeps its own axis fixed and rotates the other two, e.g.

$$R_y(\phi) = \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix}. \tag{26}$$

The camera forward axis maps to $R[1, 0, 0]^{\top} = (0.936, 0.020, -0.352)$, i.e. 20.6° below horizontal.

Step 6. Floor filter: keep $0 < z < 5$ m and $|p_{B,z} - z_{\mathrm{floor}}| < 0.05$ m, with $z_{\mathrm{floor}} = -0.14$ m measured from the data (not 0).

Step 7. World and paint: $\mathbf{p}_W = R_k\, \mathbf{p}_{B,xy} + \mathbf{t}_k$, then $c\big(\mathrm{cell}(\mathbf{p}_W)\big) \leftarrow \mathrm{RGB}(u_{\mathrm{rgb}}, v_{\mathrm{rgb}})$.

TABLE II
Trajectory quality.

Data	Method	Path	Gap to start	Cross-pass
20	odometry	95.9 m	2.77 m	18.7 cm
20	ICP	95.3 m	2.69 m	10.8 cm
20	pose graph	93.9 m	0.12 m	2.7 cm
21	odometry	91.8 m	3.27 m	17.1 cm
21	ICP	96.7 m	2.80 m	11.4 cm
21	pose graph	93.7 m	0.22 m	1.1 cm

TABLE III
Pose graph tuning, dataset 20.

Variant	Cross-pass	Path
ICP only, no pose graph	10.8 cm	95.3 m
fixed-interval closures only	11.4 cm	94.4 m
odometry $\sigma = 5$ cm, 0.02 rad	1.7 cm	86.5 m
odometry $\sigma = 1$ cm, 0.005 rad (used)	2.7 cm	93.9 m

IV. Results

ICP warm-up. The mean distance from each view point to the model is 1.2 to 3.2 mm for all eight views (Fig. 3).

Trajectories. Without ground truth, two checks are used: the robot returns to its start, so the gap to the start should be near 0; and cross-pass consistency, the median distance from a scan's points to points seen on a different pass (≥ 500 scans apart). A correct path makes both passes see the same wall in the same place.

ICP roughly halves the odometry error but still ends 2.7 m from the start (Table II, Fig. 4). The pose graph closes the loop: 0.12 and 0.22 m, with 2.7 and 1.1 cm cross-pass error. It accepted 496 fixed-interval and 121 of 875 proximity closures on dataset 20, and 475 and 233 of 832 on dataset 21. Many proximity candidates fail because the robot passes the same spot in the opposite direction, and a 270° LiDAR then sees different walls.

Fixed-interval closures alone change almost nothing, since scans 10 apart repeat what the odometry factors already state; the proximity closures remove the drift (Table III). Loose odometry noise gives the best cross-pass value but shortens the path by 9 m, so path length is checked as well.

Maps. Figs. 5 and 6 show the maps over time and ICP vs. optimized poses at the end. With ICP poses the return pass is about 11 cm off and the path end misses the start, which blurs the start area and the walls. With optimized poses both passes overlap and the hallways form a clean L of about 25 m by 22 m. Run time: scan matching 43 s, pose graph 30 s, occupancy map 70 s, texture map 20 s.

Discussion. Odometry drifts from wheel slip and gyro bias; ICP uses the walls but only compares nearby scans, so it still drifts; the pose graph links scans far apart in time, which removes the drift. Three early failures and their fixes: matching to the previous scan shrank the hallways (keyframes), loose odometry noise bent the

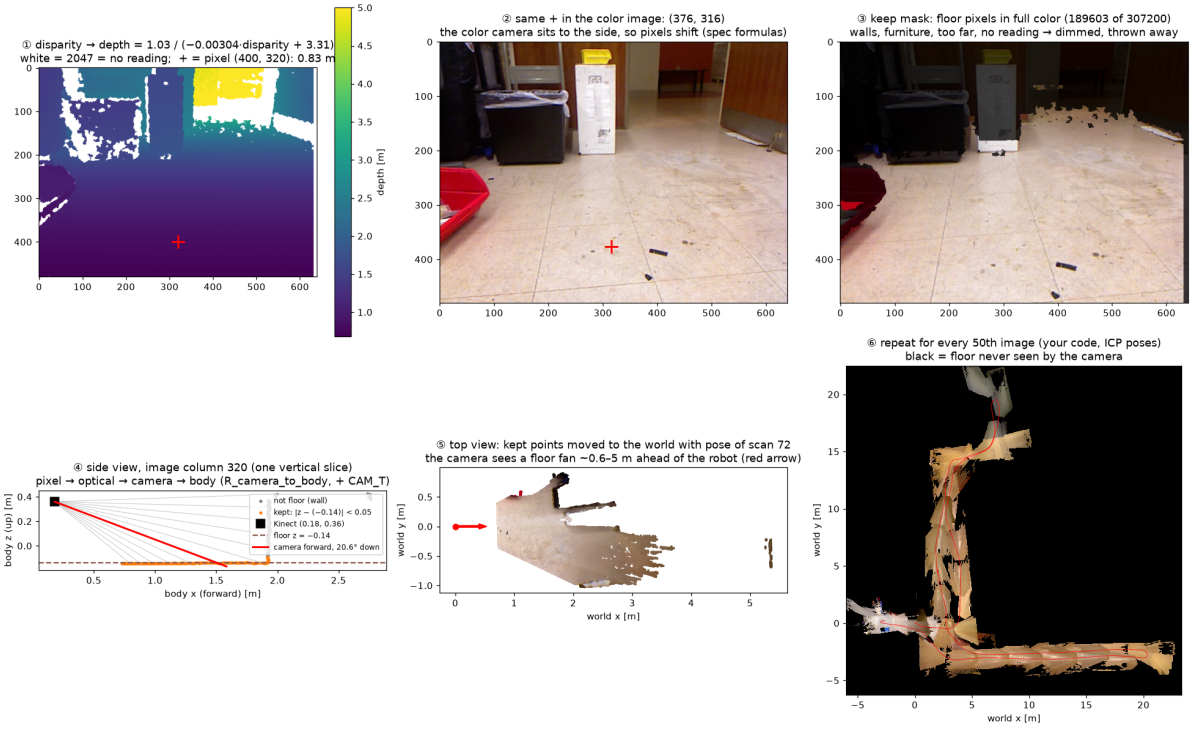


Fig. 2. One Kinect image through the texture pipeline (dataset 20, image 1): (1) depth from disparity; (2) the same pixel in the color image; (3) kept floor pixels; (4) side view of one column: rays from the tilted camera hit the floor at $z = -0.14\,\mathrm{m}$; (5) kept points in the world; (6) all images painted.

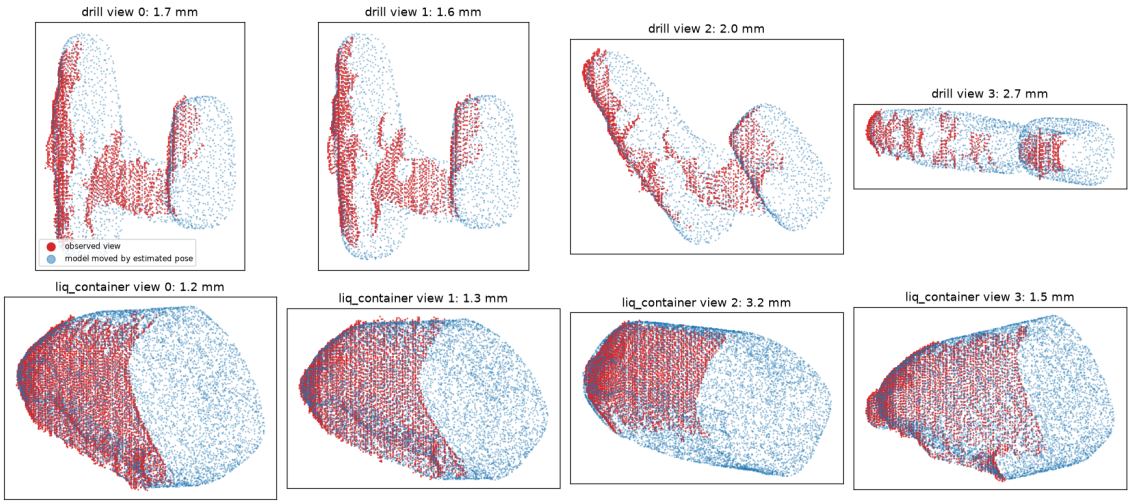


Fig. 3. ICP warm-up, top view. Red: partial view. Blue: model moved by the estimated pose. Title: mean point distance.

map (small σ), and ICP from a bad start locked onto a wrong pose (12 initial yaws). The global rotation of the map depends on the first hallway and cannot be checked without ground truth.

Acknowledgment

Claude (an AI assistant) was used as a tutor for hints and code checks during the project, and to help draft this report from the results.

References

[1] N. Atanasov, "ECE 276A: Sensing & Estimation in Robotics," UC San Diego. [Online]. Available: https://natanaso.github.io/ece276a/

[2] "Robot configuration," ECE 276A Project 2 handout (RobotConfiguration.pdf).

[3] F. Dellaert and GTSAM contributors, "GTSAM." [Online]. Available: https://github.com/borglab/gtsam

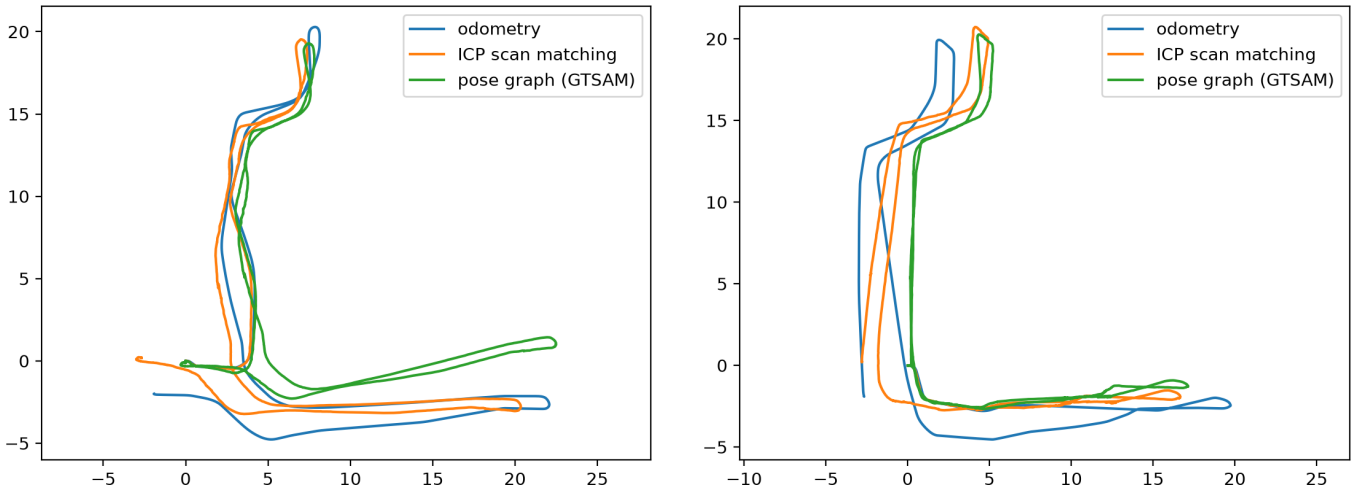


Fig. 4. Trajectories in meters: dataset 20 (left) and 21 (right).

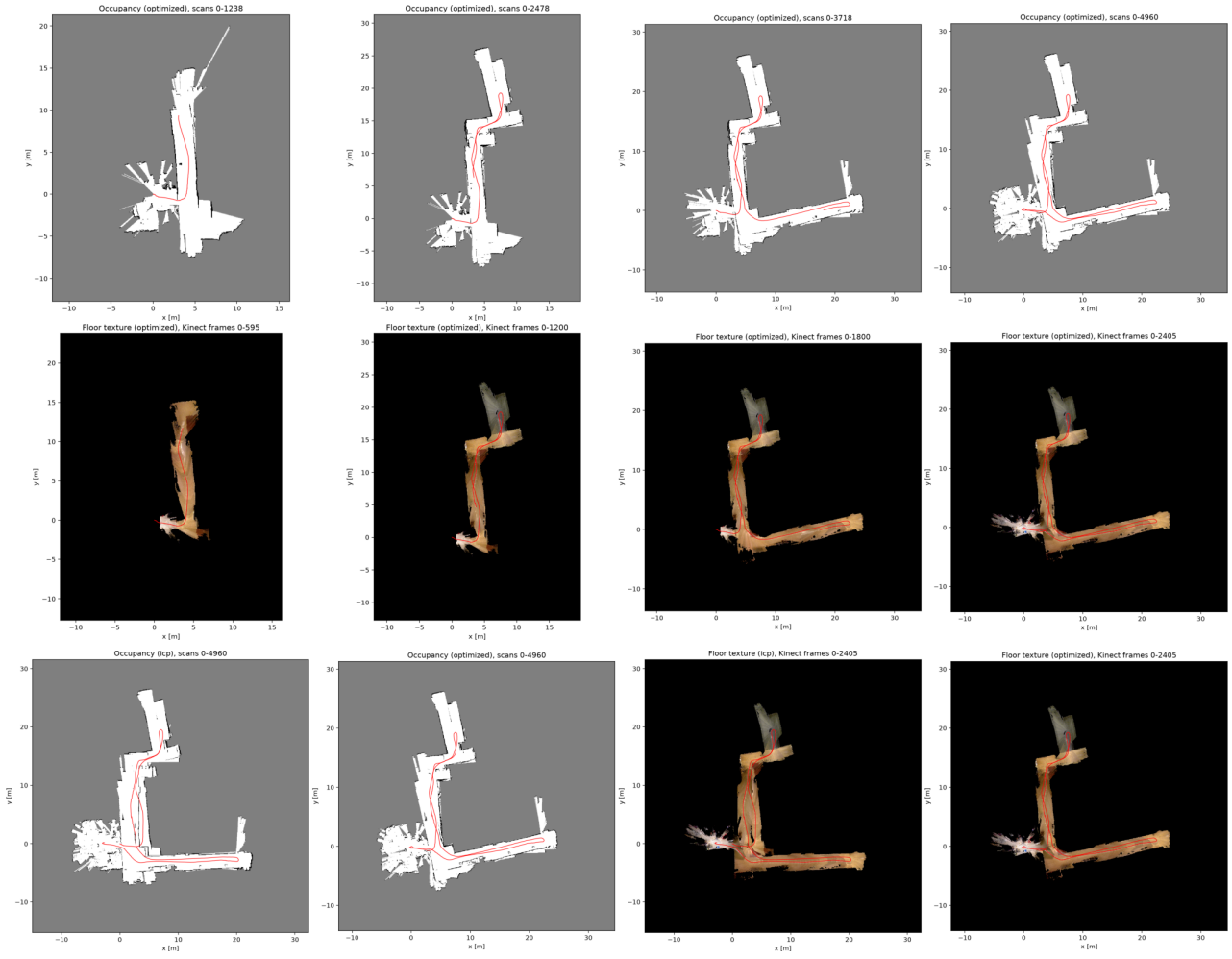


Fig. 5. Dataset 20. Top: trajectory and occupancy map over time (optimized poses; black = wall, white = free, gray = unknown, red = path so far). Middle: floor texture over time. Bottom: final maps, ICP vs. optimized poses.

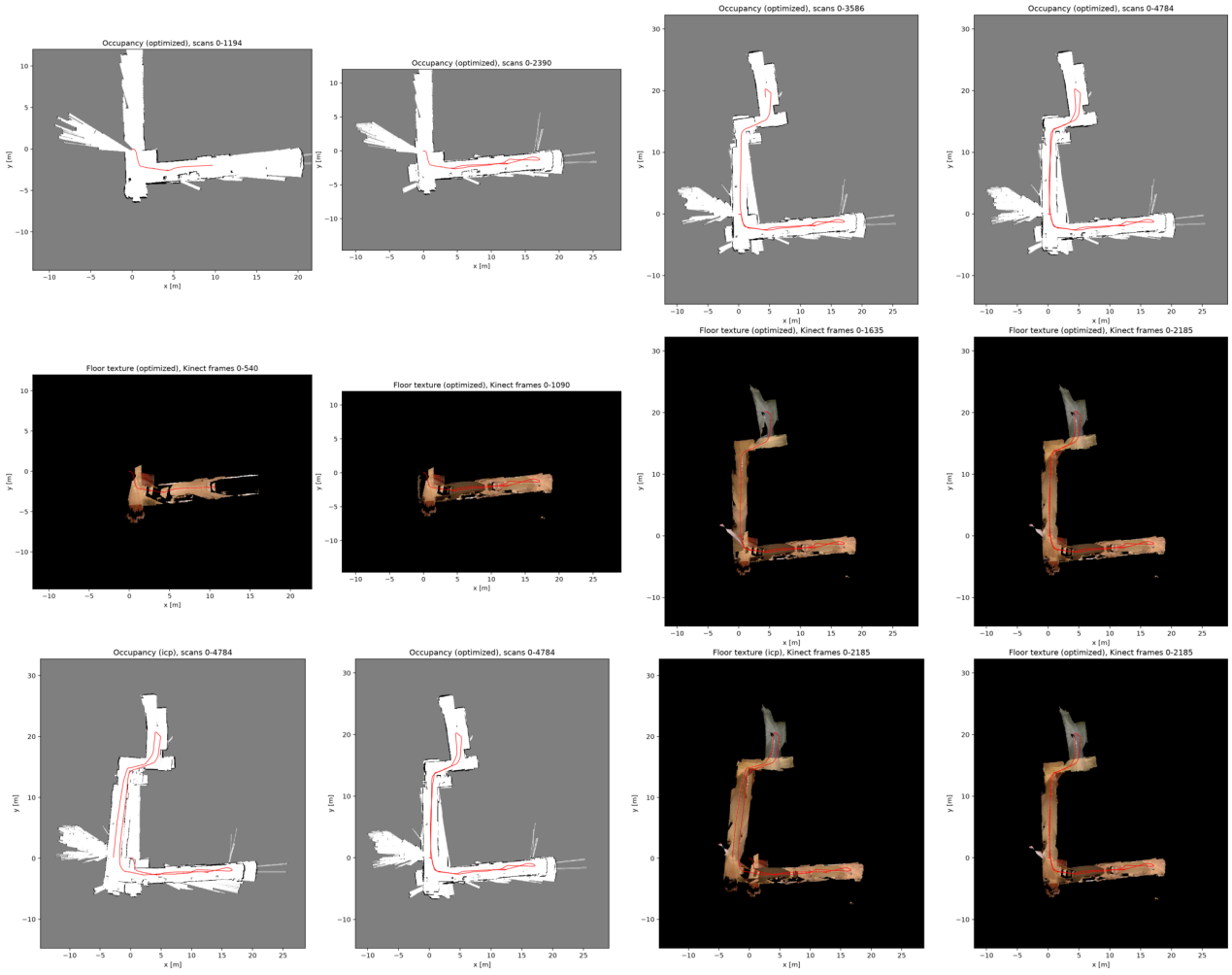


Fig. 6. Dataset 21, same layout as Fig. 5.